

Cancer Genomics & Bioinformatics

Day 1: Database mining

Aristeidis G. Telonis, PhD

Research Assistant Professor

Department of Biochemistry & Molecular Biology

Sylvester Comprehensive Cancer Center

University of Miami Miller school of Medicine

<https://github.com/atelonis/BioinformaticsExercises>



Learning objectives

Databases:

- How is biological information organized in databases?
- What are some commonly used biological databases?
- How can I access genomic data?

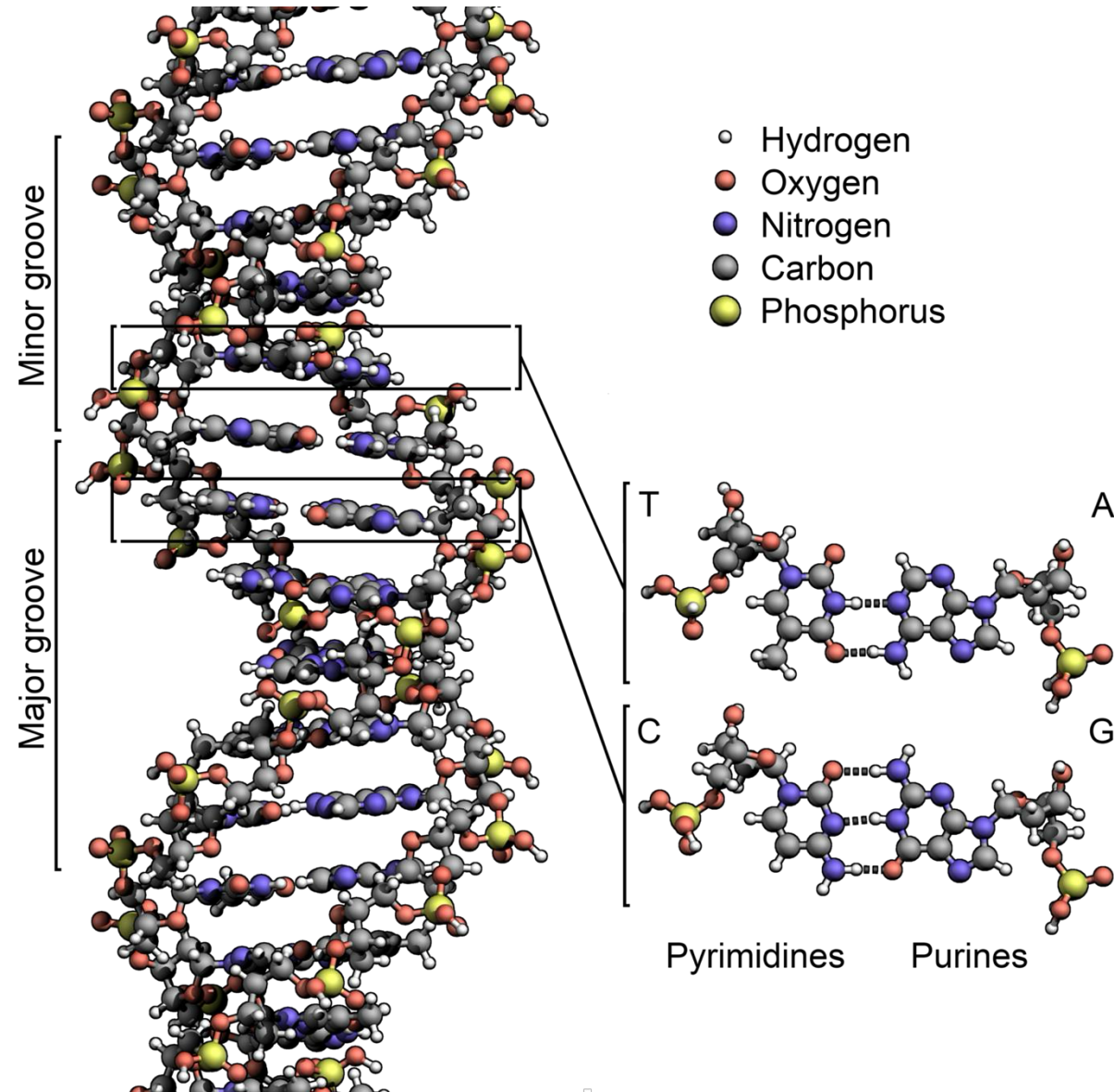
How do I use linux to answer questions like:

- Which chromosome has the most coding genes?
- Which coding gene has the most coding splice variants?
- Which coding gene has the most non-coding variants?

@SylvesterCancer



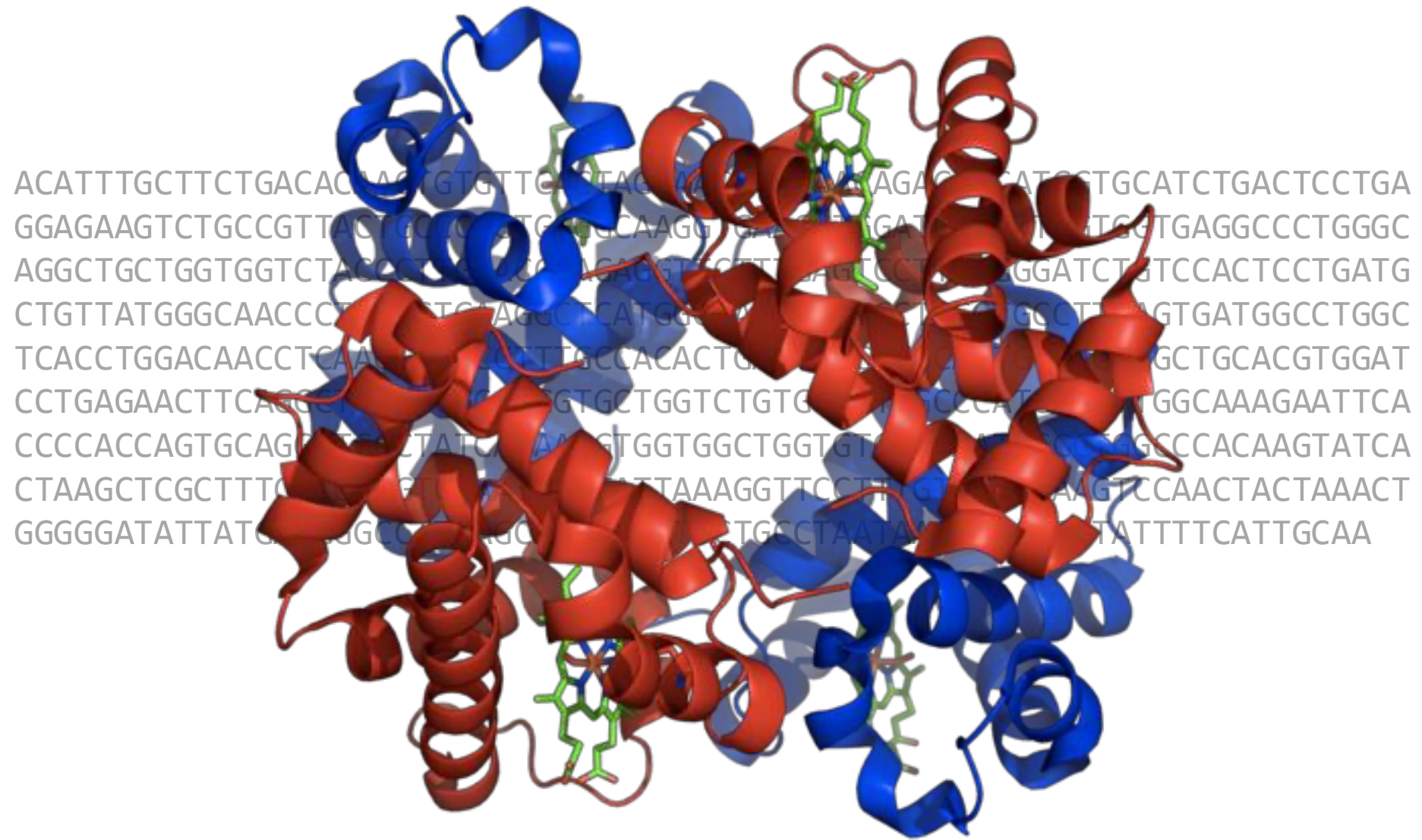
Genome organization: from biology to computer bytes



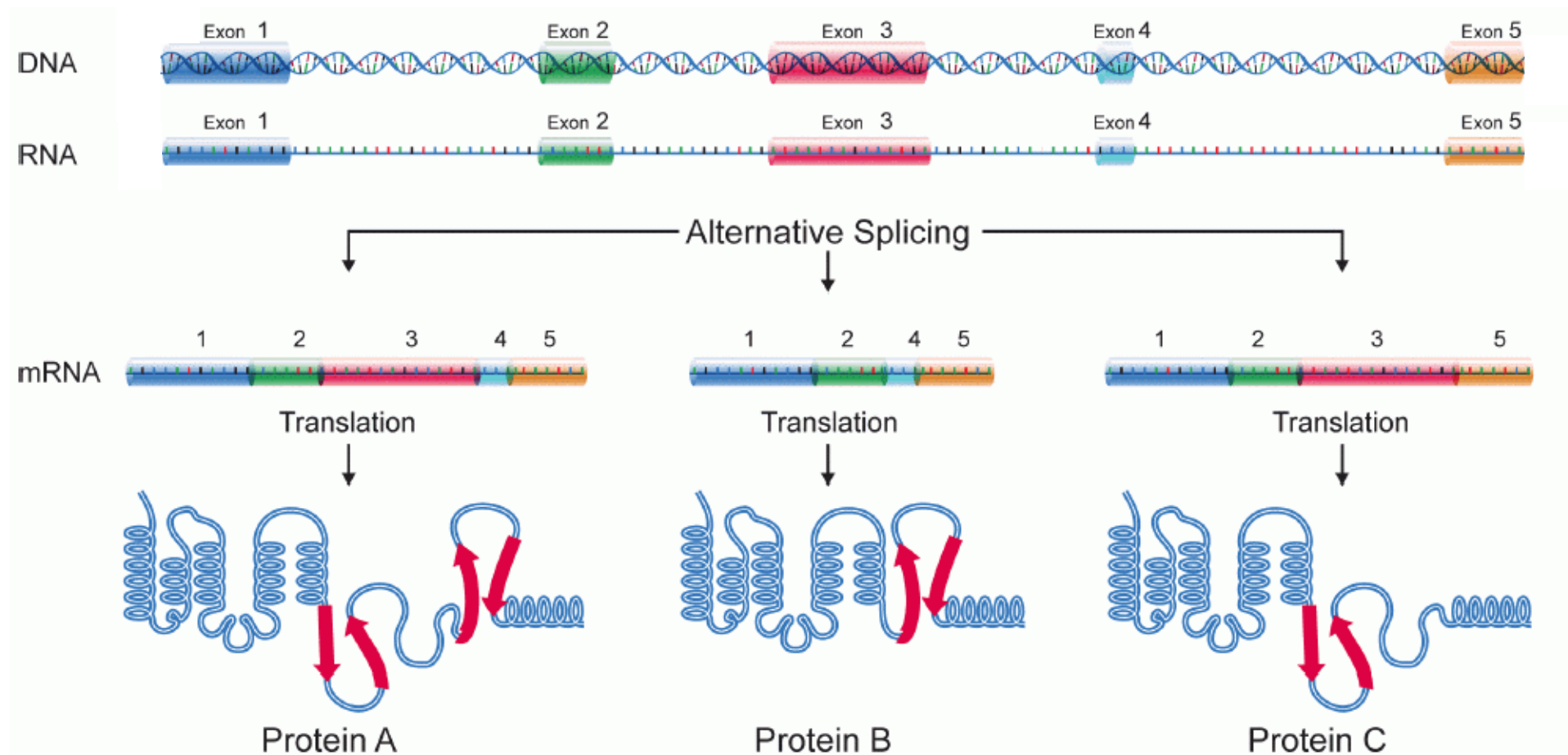
ACATTTGCTTCTGACACAACCTGTGTTCACTAGCAACCTCAAACAGACACCATGGTGCATCTGACTCCTGA
GGAGAAGTCTGCCGTTACTGCCCTGTGGGGCAAGGTGAACGTGGATGAAGTTGGTGGTGAGGCCCTGGGC
AGGCTGCTGGTGGTCTACCCTTGGACCCAGAGGTTCTTTGAGTCCTTTGGGGATCTGTCCACTCCTGATG
CTGTTATGGGCAACCCTAAGGTGAAGGCTCATGGCAAGAAAGTGCTCGGTGCCTTTAGTGATGGCCTGGC
TCACCTGGACAACCTCAAGGGCACCTTTGCCACACTGAGTGAGCTGCACTGTGACAAGCTGCACGTGGAT
CCTGAGAACTTCAGGCTCCTGGGCAACGTGCTGGTCTGTGTGCTGGCCCATCACTTTGGCAAAGAATTCA
CCCCACCAGTGCAGGCTGCCTATCAGAAAGTGGTGGCTGGTGTGGCTAATGCCCTGGCCCACAAGTATCA
CTAAGCTCGCTTTCTTGCTGTCCAATTTCTATTAAAGGTTCCCTTTGTTCCCTAAGTCCAACCTACTAACT
GGGGGATATTATGAAGGGCCTTGAGCATCTGGATTCTGCCTAATAAAAAACATTTATTTTCATTGCAA

+ 6.4 billion letters

DNA is a sequence of four letters



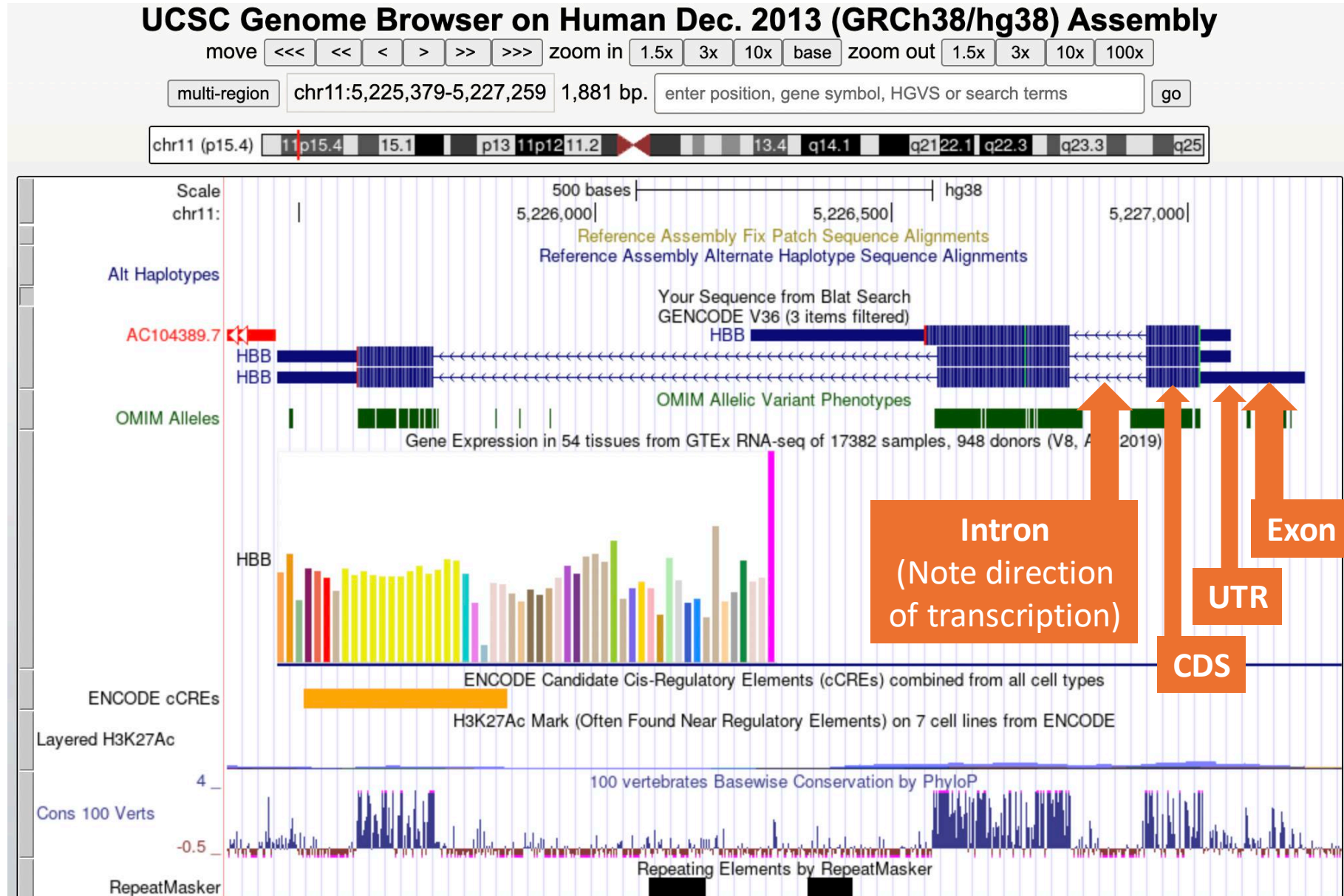
We need to account for the complexity of molecular biology



By National Human Genome Research Institute
http://www.genome.gov/Images/EdKit/bio2j_large.gif, Public Domain
<https://commons.wikimedia.org/w/index.php?curid=2132737>

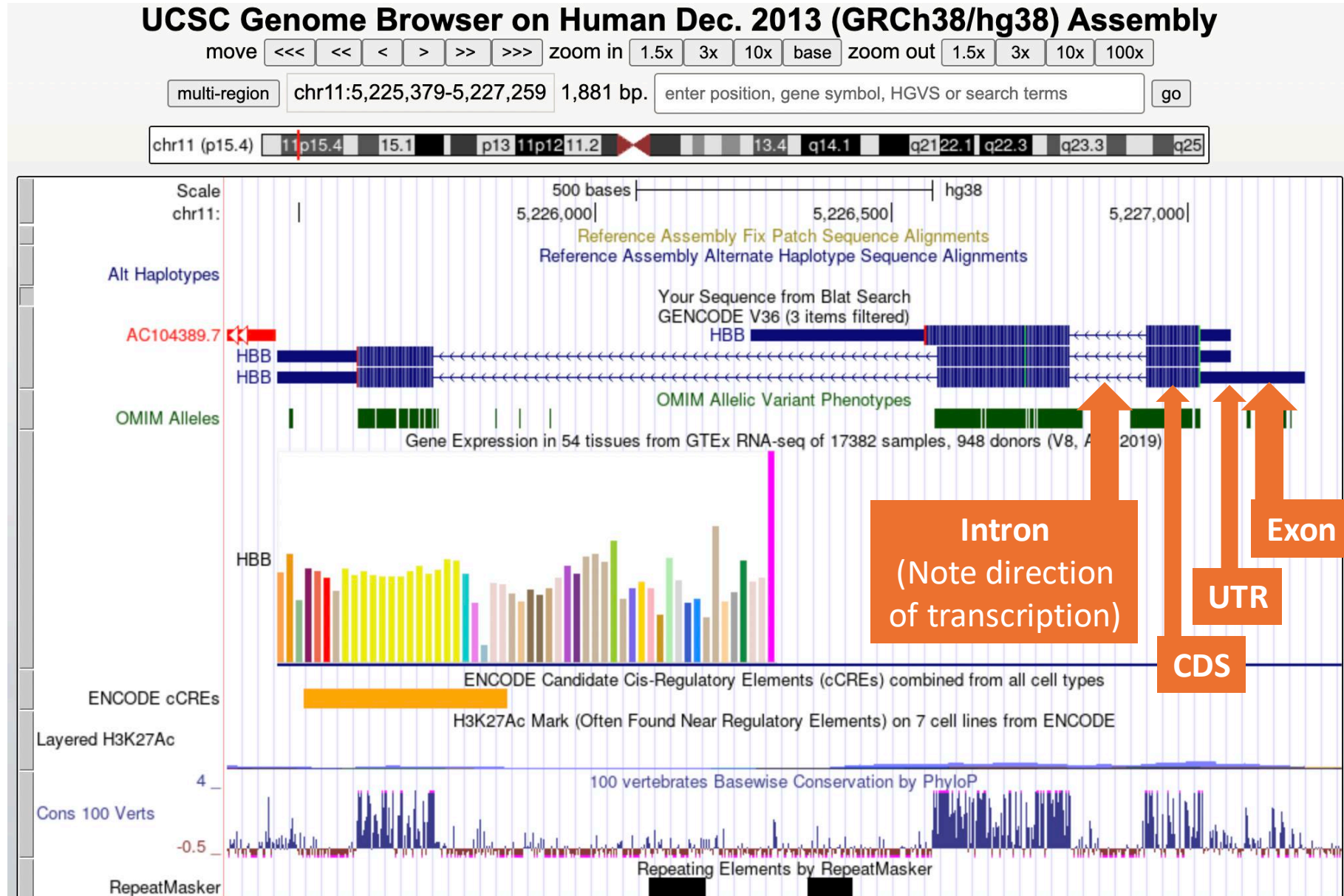
The above mRNAs are protein-coding transcripts. There are a lot of transcripts that are not translated into proteins and are therefore non-coding

The UCSC Genome Browser: view of HBB (Hemoglobin beta)



http://genome.ucsc.edu/cgi-bin/hgTracks?db=hg38&lastVirtModeType=default&lastVirtModeExtraState=&virtModeType=default&virtMode=0&nonVirtPosition=&position=chr11%3A5225379%2D5227259&hgid=1106581317_2aa2ojxb5YFYRt0jEidpa8

The UCSC Genome Browser: view of HBB (Hemoglobin beta)



http://genome.ucsc.edu/cgi-bin/hgTracks?db=hg38&lastVirtModeType=default&lastVirtModeExtraState=&virtModeType=default&virtMode=0&nonVirtPosition=&position=chr11%3A5225379%2D5227259&hgid=1106581317_2aa2ojxb5YFYRt0jEidpa8

- Feel free to explore the databases:

UCSC

http://genome.ucsc.edu/cgi-bin/hgGene?hgg_gene=ENST00000335295.4&hgg_chrom=chr11&hgg_start=5225463&hgg_end=5227071&hgg_type=knownGene&db=hg38

ENSEMBL

http://useast.ensembl.org/Homo_sapiens/Gene/Summary?db=core;g=ENSG00000244734;r=11:5225464-5229395

NCBI

https://www.ncbi.nlm.nih.gov/gene?cmd=retrieve&dopt=default&rn=1&list_uids=3043

We know the genome sequence

We know the protein coding genes (location, size, exons, introns)

We know the non-coding genes

→ http://useast.ensembl.org/Homo_sapiens/Info/Annotation

Gene counts (Primary assembly)

Coding genes	19,871 (excl 661 readthrough)
Non coding genes	42,126
Small non coding genes	4,866
Long non coding genes	35,044 (excl 301 readthrough)
Misc non coding genes	2,216
Pseudogenes	15,198 (excl 1 readthrough)
Gene transcripts	387,954

http://www.ensembl.org/Homo_sapiens/Info/Annotation

 An official website of the United States government [Here's how you know](#) ▾

 **National Library of Medicine**
National Center for Biotechnology Information

Log in

Gene

Gene ▾

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HBB hemoglobin subunit beta [*Homo sapiens* (human)]
Gene ID: 3043, updated on 3-May-2025

Summary

Official Symbol

HBB

provided by [HGNC](#)

Official Full Name

hemoglobin subunit beta

provided by [HGNC](#)

Primary source

[HGNC:HGNC:4827](#)

See related

[Ensembl:ENSG00000244734](#) [MIM:141900](#) [AllianceGenome:HGNC:4827](#)

Gene type

protein coding

RefSeq status

REVIEWED

Organism

[Homo sapiens](#)

Lineage

Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Eutheria; Homo

Also known as

ECYT6; CD113t-C; beta-globin

Summary

The alpha (HBA) and beta (HBB) loci determine the structure of the 2 types of polypeptide chains that form the hemoglobin tetramer consists of two alpha chains and two beta chains. Mutant beta globin causes beta-thalassemia. Reduced amounts of detectable beta globin causes beta-plus-thalassemia. gamma-G -- gamma-A -- delta -- beta--3'. [provided by RefSeq, Jul 2008]

Orthologs

[all](#)

NEW

Try the new [Gene table](#)

Try the new [Transcript table](#)

Genomic context

Location:

11p15.4

Exon count:

3

Table of contents

[Go to the HIV-1 Human Interaction Database](#)

Pathways from PubChem

14 items

Download

Transport of small molecules

Data Source: [Reactome](#) External ID: [R-HSA-382551](#)

Taxonomy Name: [Homo sapiens \(human\)](#)

O2/CO2 exchange in erythrocytes

Data Source: [Reactome](#) External ID: [R-HSA-1480926](#)

Taxonomy Name: [Homo sapiens \(human\)](#)

Erythrocytes take up carbon dioxide and release oxygen

Data Source: [Reactome](#) External ID: [R-HSA-1237044](#)

Taxonomy Name: [Homo sapiens \(human\)](#)

Erythrocytes take up oxygen and release carbon dioxide

Data Source: [Reactome](#) External ID: [R-HSA-1247673](#)

Taxonomy Name: [Homo sapiens \(human\)](#)

Vesicle-mediated transport

Data Source: [Reactome](#) External ID: [R-HSA-5653656](#)

Taxonomy Name: [Homo sapiens \(human\)](#)

Binding and Uptake of Ligands by Scavenger Receptors

Data Source: [Reactome](#) External ID: [R-HSA-2173782](#)

Taxonomy Name: [Homo sapiens \(human\)](#)

Scavenging of heme from plasma

Data Source: [Reactome](#) External ID: [R-HSA-2168880](#)

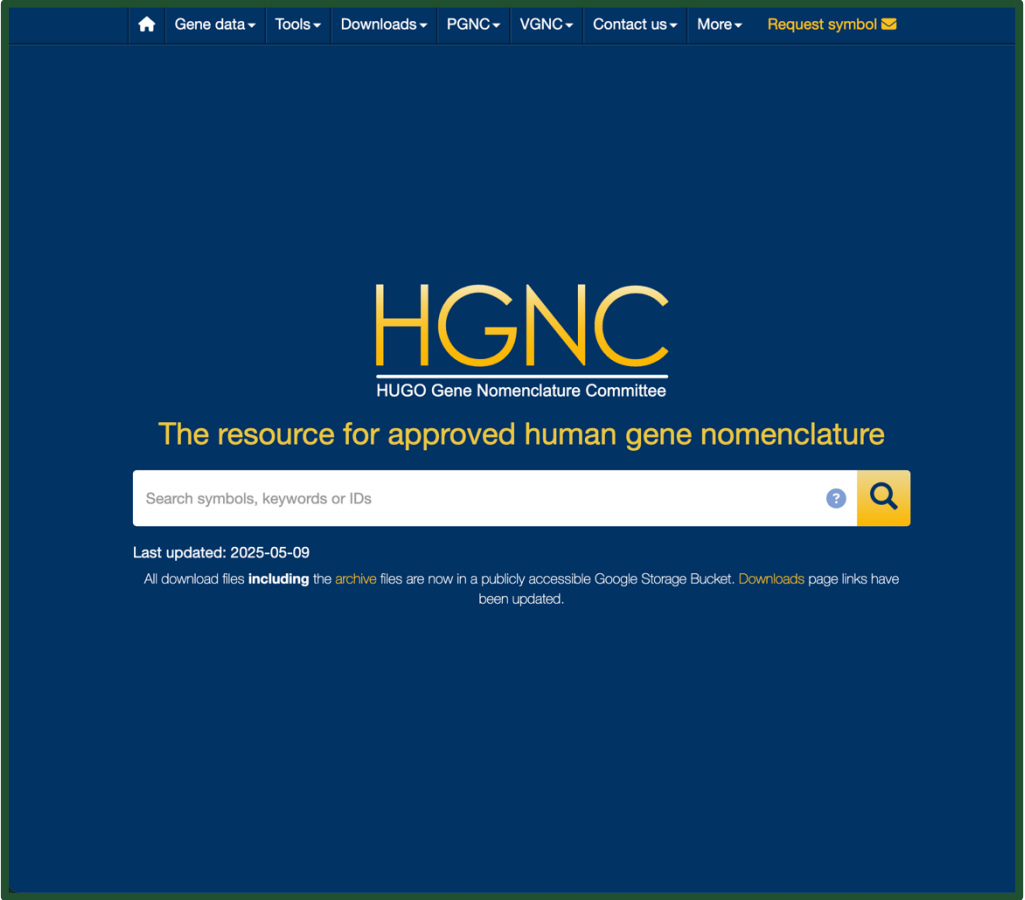
Taxonomy Name: [Homo sapiens \(human\)](#)

Hemostasis

Data Source: [Reactome](#) External ID: [R-HSA-109582](#)

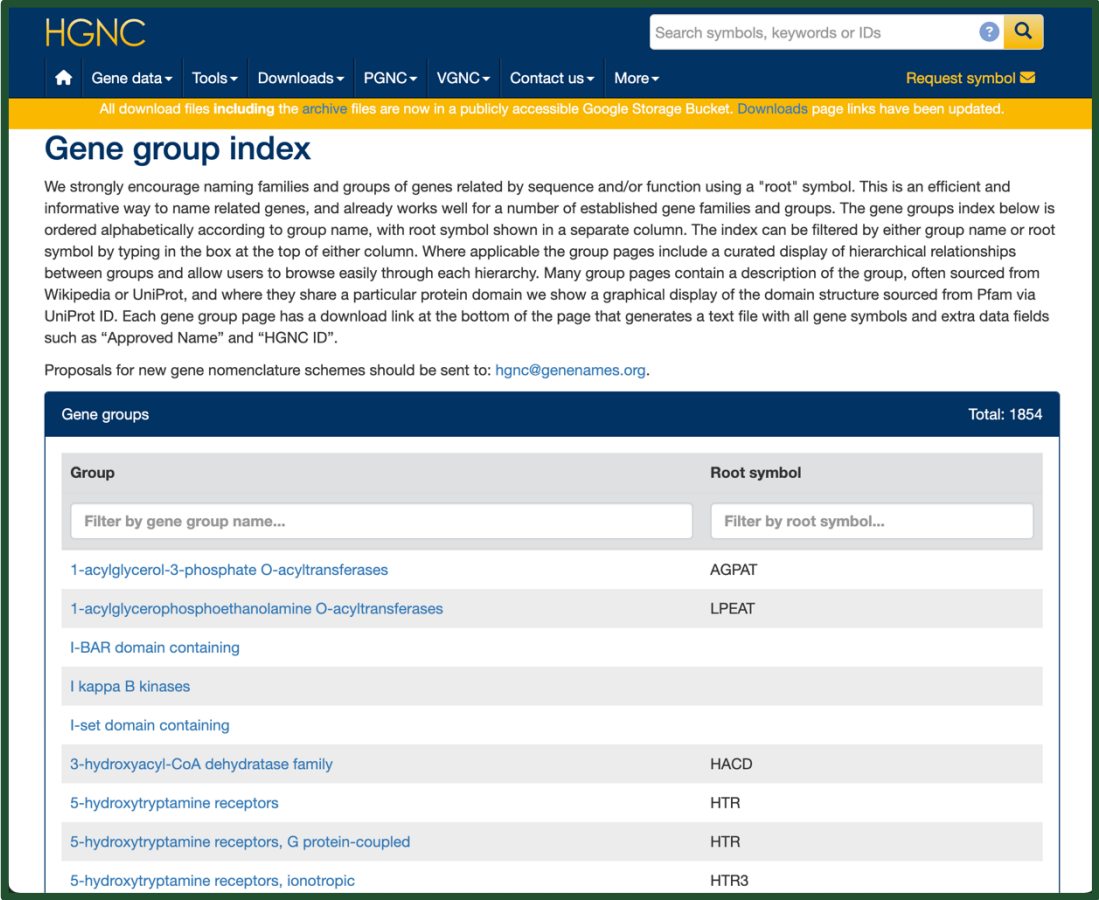
Taxonomy Name: [Homo sapiens \(human\)](#)

<https://www.ncbi.nlm.nih.gov/gene/3043>



<https://www.genenames.org/>

Gene data > Gene group reports



Gene groups

Total: 8

Group

Root symbol

acetylase

Filter by root symbol...

Arylacetamide deacetylase family

AADAC

Histone deacetylase superfamily

HDAC, SIRT

Histone deacetylases, class IIA

HDAC

Histone deacetylases, class IIB

HDAC

Histone deacetylases, class I

HDAC

Histone deacetylases, class II

HDAC

Histone deacetylases, class IV

HDAC

SIN3 histone deacetylase complex subunits

10

25

50

100

Gene group: Histone deacetylases, class I (HDAC) ?

A subgroup of ? : "Histone deacetylase superfamily (HDAC, SIRT)"

Gene group hierarchy map ?

Rearrange mode

Histone deacetylase superfamily

Histone deacetylases, class I

The mapped domains of Q13547, encoded by the HDAC1 gene, an example gene within the group. [Source: Pfam & UniProt]

Gene: HDAC1, UniProt: Q13547

Histone deacetylase domain

482 amino acids

Genes contained within the group: 4 ?

HGNC ID (gene)	Approved symbol	Approved name	Previous symbols	Aliases	Chromosome
HGNC:4852	HDAC1	histone deacetylase 1	RPD3L1	HD1, GON-10, KDAC1	1p35.2-p35.1
HGNC:4853	HDAC2	histone deacetylase 2		RPD3, YAF1, KDAC2	6q21
HGNC:4854	HDAC3	histone deacetylase 3		RPD3, HD3, RPD3-2, KDAC3	5q31.3
HGNC:13315	HDAC8	histone deacetylase 8	HDACL1, WTS, MRXS6	RPD3, KDAC8	Xq13.1

Downloads ?

Genes within the current group only:

TXTJSON

back to index ...



HumanMouseHow to access dataFAQDocumentationAbout us

Human

More about GENCODE Human

Statistics about the GENCODE Release 48

The statistics derive from the [gtf file](#) that contains only the annotation of the main chromosomes.

For details about the calculation of these statistics please see the [README_stats.txt file](#).

General stats

Total No of Genes	78686	Total No of Transcripts	385669
Protein-coding genes	19435	Protein-coding transcripts	89843
- readthrough genes (not included)	661	- full length protein-coding	65024
Long non-coding RNA genes	35901	- partial length protein-coding	24819
Small non-coding RNA genes	7563	Nonsense mediated decay transcripts	21902
Pseudogenes	14695	Long non-coding RNA loci transcripts	191076
- processed pseudogenes	10643		
- unprocessed pseudogenes	3549		
- unitary pseudogenes	266		
Immunoglobulin/T-cell receptor gene segments		Total No of distinct translations	65814
- protein coding segments	412	Genes that have more than one distinct translations	13648
- pseudogenes	237		

<https://www.genencodegenes.org/>



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The GENCODE Project: Encyclopædia of genes and gene variants

Background

The National Human Genome Research Institute (NHGRI) launched a public research consortium named [ENCODE](#), the Encyclopedia Of DNA Elements, in September 2003, to carry out a project to identify all functional elements in the human genome sequence. After a successful pilot phase on 1% of the genome, the scale-up to the entire genome is now underway. The Wellcome Sanger Institute was [awarded a grant](#) to carry out a scale-up of the GENCODE project for integrated annotation of gene features.

Having been involved in successfully delivering the definitive annotation of functional elements in the human genome, the GENCODE group were [awarded a second grant](#) in 2013 in order to continue their human genome annotation work and expand GENCODE to include annotation of the mouse genome. A [third grant](#) was awarded in 2017 for the continued improvement of the annotation of the human and mouse genomes, and a fourth grant followed in 2021. Details of the grants awarded can be found [here](#).

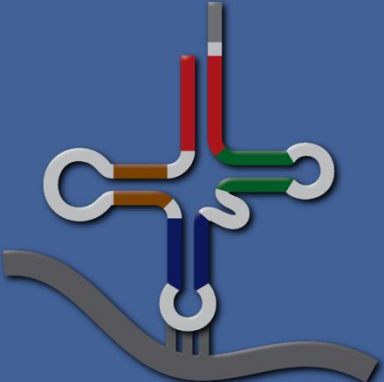
GENCODE gene annotation is used by a large number of international consortia.

[Home](#)

GtRNAdb

tRNAscan-SE analysis of complete genomes

The genomic tRNA database contains tRNA gene predictions made by **tRNAscan-SE** on complete or nearly complete genomes. Unless otherwise noted, all annotation is automated, and has not been inspected for agreement with published literature.



Data Release 22 (Sept 2024)

To access the previous releases of GtRNAdb or check out the change log, please click [here](#)

Links to Most Viewed Genomes

<i>Homo sapiens</i> (GRCh37/hg19)	<i>Homo sapiens</i> (GRCh38/hg38)
<i>Mus musculus</i> (GRCm38/mm10)	<i>Mus musculus</i> (GRCm39/mm39)
<i>Saccharomyces cerevisiae</i> S288c	<i>Schizosaccharomyces pombe</i> 972h-
<i>Caenorhabditis elegans</i>	<i>Drosophila melanogaster</i> (BDGP Rel.

Additional tRNA Tools

tDRnamer - standardized naming for tRNA-derived RNAs

tRAX - tRNA and tRNA-derived RNA sequencing data analysis

<https://gttnadb.ucsc.edu/>

[miRBase](#)

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miRBase: the microRNA database

the archive for microRNA sequences and annotations

[More about miRBase →](#)



By identifier or keyword



By genomic location



Browse



By tissue expression



By sequence



Downloads



Comments or questions? Email mirbase@manchester.ac.uk

<https://www.mirbase.org/>

UniProt

BLASTAlignPeptide searchID mappingSPARQL

Release 2025_02 | Statistics

Help

Find your protein

UniProtKB

Examples: Insulin, APP, Human, P05067, organism_id:9606

UniProt is the world's leading high-quality, comprehensive and freely available protein knowledgebase

Proteins
UniProt Knowledgebase

Species
Proteomes

<https://www.uniprot.org/>

UniProt

BLASTAlignPeptide searchID mappingSPARQL

UniProtKB

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Search

Help

Q12888 • TP53B_HUMAN

Function

Names & Taxonomy

Subcellular Location

Disease & Variants

PTM/Processing

Expression

Interaction

Structure

Family & Domains

Sequence & Isoforms

Similar Proteins

Proteinⁱ

Geneⁱ

Statusⁱ

Organismⁱ

TP53-binding protein 1

TP53BP1

UniProtKB reviewed (Swiss-Prot)

Homo sapiens (Human)

Amino acids

Protein existenceⁱ

Annotation scoreⁱ

1972 (go to sequence)

Evidence at protein level

5/5

Entry

Variant viewer2,324

Feature viewer

Genomic coordinates

Publications

External links

History

Tools

Download

Add

Add a publication

Entry feedback

Functionⁱ

Double-strand break (DSB) repair protein involved in response to DNA damage, telomere dynamics and class-switch recombination (CSR) during antibody genesis (PubMed:12364621, PubMed:17190600, PubMed:21144835, PubMed:22553214, PubMed:23333306, PubMed:27153538, PubMed:28241136, PubMed:31135337, PubMed:37696958). Plays a key role in the repair of double-strand DNA breaks (DSBs) in response to DNA damage by promoting non-homologous end joining (NHEJ)-mediated repair of DSBs and specifically counteracting the function of the homologous recombination (HR) repair protein BRCA1 (PubMed:22553214, PubMed:23333306, PubMed:23727112, PubMed:27153538, PubMed:31135337). In response to DSBs, phosphorylation by ATM promotes interaction with RIF1 and dissociation from NUDT16L1/TIRR, leading to recruitment to DSBs sites (PubMed:28241136). Recruited to DSBs sites by recognizing and binding histone H2A monoubiquitinated at 'Lys-15' (H2AK15Ub) and histone H4

PATHWAYS | KEGG as a reference database for metabolic, signaling and organismal processes

KEGG

Databases

Tools

Auto annotation

Kanehisa Lab



KEGG PATHWAY Database

Wiring diagrams of molecular interactions

KEGG2 PATHWAY BRITE MODULE KO GENES

Select prefix

map

Organism

Enter keywords

Pathway Maps

KEGG PATHWAY is a collection of manually drawn pathway reaction and relation networks for:

1. Metabolism

Global/overview Carbohydrate Energy Lipid Nucleic acid
Cofactor/vitamin Terpenoid/PK Other secondary metabolites

2. Genetic Information Processing

3. Environmental Information Processing

4. Cellular Processes

5. Organismal Systems

6. Human Diseases

7. Drug Development

The pathway map viewer linked from this page is a part of

Pathway Identifiers

Each pathway map is identified by the combination of 2-4 letters and numbers having the following meaning:

map manually drawn reference pathway
ko reference pathway highlighting KOs
ec reference metabolic pathway highlighting EC numbers
rn reference metabolic pathway highlighting reaction numbers
<org> organism-specific pathway generated by conversion of KOs to organismal pathways
vg viruses pathway generated by converting KOs to viral pathways
vx viruses extended pathway generated by including host pathways

and the numbers starting with the following:

011 global map (lines linked to KOs)
012 overview map (lines linked to KOs)
013 higher (multi-organism) level overview map (lines linked to KOs)
010 chemical structure map (no KO expansion)
07 drug structure map (no KO expansion)
other regular map (boxes linked to KOs)

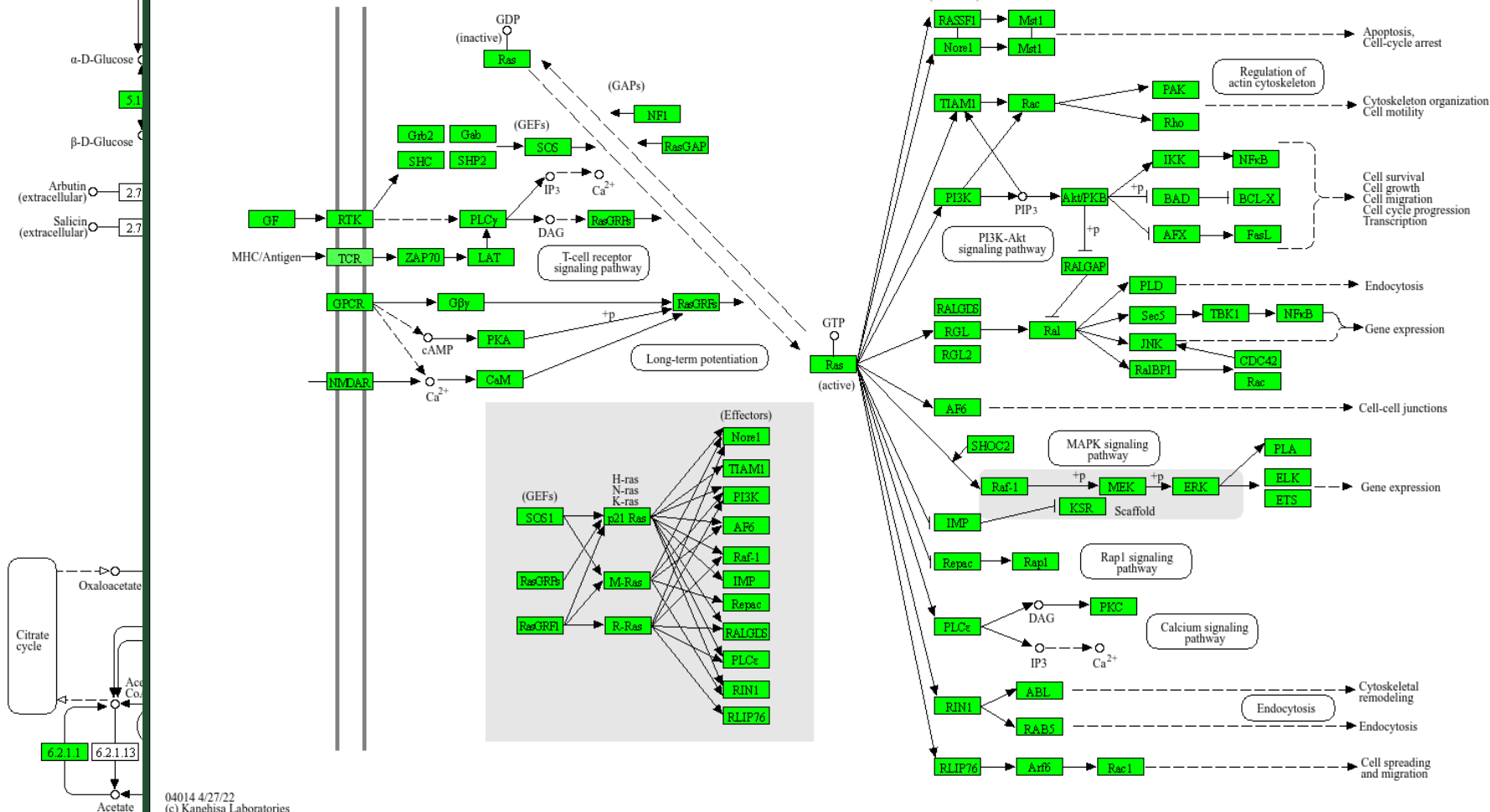
are used for different types of maps.

<https://www.kegg.jp/>

GLYCOLYSIS / GLUCONEOGENESIS

Starch and sucrose

RAS SIGNALING PATHWAY



00010 6/10/24
(c) Kanehisa Laboratories

The Molecular Signatures Database (MSigDB) is a resource of tens of thousands of annotated gene sets for use with GSEA software, divided into [Human](#) and [Mouse](#) collections. From this web site, you can

- ▶ **Examine** a gene set and its annotations. See, for example, the [HALLMARK_APOPTOSIS human gene set page](#).
- ▶ **Browse** gene sets by name or collection.
- ▶ **Search** for gene sets by keyword.
- ▶ **Investigate** gene sets:
 - ▶ **Compute overlaps** between your gene set and gene sets in MSigDB.
 - ▶ **Categorize** members of a gene set by gene families.
 - ▶ **View the expression profile** of a gene set in a provided public expression compendia.
 - ▶ Investigate the gene set in the online **biological network repository NDEx**
- ▶ **Download** gene sets.

GSEA and MSigDB are available for use under [these license terms](#).

Please [register](#) to download the GSEA software and the MSigDB gene sets, and to use our web tools. After registering, you can log in at any time using your email address. Registration is free. Its only purpose is to help us track usage for reports to our funding agencies.

Human Collections

H **hallmark gene sets** are coherently expressed signatures derived by aggregating many MSigDB gene sets to represent well-defined biological states or processes.

C1 **positional gene sets** corresponding to human chromosome cytogenetic bands.

C2 **curated gene sets** from online pathway databases, publications in PubMed, and knowledge of domain experts.

C3 **Regulatory target gene sets** based on gene target predictions for microRNA seed sequences and predicted transcription factor binding sites.

C4 **computational gene sets** defined by mining large collections of cancer-oriented expression data.

MH **mouse-ortholog hallmark gene sets** are versions of gene sets in the MSigDB Hallmarks collection mapped to their mouse orthologs.

- ▶ **CP:KEGG_LEGACY** (KEGG Legacy gene sets, 186 gene sets)

BECKER_TAMOXIFEN_RESISTANCE_DN	IIZUKA_LIVER_CANCER_PROGRESSION_L0_L1_DN	PYEON_CANCER_HEAD_AND_NECK_V5_CERVICAL_DN
BECKER_TAMOXIFEN_RESISTANCE_UP	IIZUKA_LIVER_CANCER_PROGRESSION_L0_L1_UP	PYEON_CANCER_HEAD_AND_NECK_V5_CERVICAL_UP
BEGUM_TARGETS_OF_PAX3_FOXP01_FUSION_DN	IIZUKA_LIVER_CANCER_PROGRESSION_L1_G1_DN	PYEON_HPV_POSITIVE_TUMORS_DN
BEGUM_TARGETS_OF_PAX3_FOXP01_FUSION_UP	IIZUKA_LIVER_CANCER_PROGRESSION_L1_G1_UP	PYEON_HPV_POSITIVE_TUMORS_UP
BEIER_GLIOMA_STEM_CELL_DN	IKEDA_MIR133_TARGETS_DN	QI_HYPOXIA
BEIER_GLIOMA_STEM_CELL_UP	IKEDA_MIR133_TARGETS_UP	QI_HYPOXIA_TARGETS_OF_HIF1A_AND_FOXA2
BENITEZ_GBM_PROTEASOME_INHIBITION_RESPONSE	IKEDA_MIR1_TARGETS_DN	QI_PLASMACYTOMA_DN
BENNETT_SYSTEMIC_LUPUS_ERYTHEMATOSUS	IKEDA_MIR1_TARGETS_UP	QI_PLASMACYTOMA_UP
S	IKEDA_MIR30_TARGETS_DN	QUINTENS_EMBRYONIC_BRAIN_RESPONSE_T0_IR
BENPORATH_CYCLING_GENES	IKEDA_MIR30_TARGETS_UP	RADA_EVA_RESPONSE_TO_IFNA1_DN
BENPORATH_EED_TARGETS	IM_SREBF1A_TARGETS	RADA_EVA_RESPONSE_TO_IFNA1_UP
BENPORATH_ES_1	INAMURA_LUNG_CANCER_SCC_DN	RADMACHER_AML_PROGNOSIS
BENPORATH_ES_2	INAMURA_LUNG_CANCER_SCC_SUBTYPES_UP	RAFFEL_VEGFA_TARGETS_DN
BENPORATH_ES_CORE_NINE	INAMURA_LUNG_CANCER_SCC_UP	RAFFEL_VEGFA_TARGETS_UP
BENPORATH_ES_CORE_NINE_CORRELATED	INGA_TP53_TARGETS	RAGHAVACHARI_PLATELET_SPECIFIC_GENE5
BENPORATH_ES_WITH_H3K27ME3	INGRAM_SHH_TARGETS_DN	
BENPORATH_MYC_MAX_TARGETS	INGRAM_SHH_TARGETS_UP	RAHMAN_TP53_TARGETS_PHOSPHORYLATED
BENPORATH_MYC_TARGETS_WITH_EBOX	IRITANI_MAD1_TARGETS_DN	RAMALHO_STEMNESS_DN
BENPORATH_NANOG_TARGETS	IRITANI_MAD1_TARGETS_UP	RAMALHO_STEMNESS_UP
BENPORATH_NOS_TARGETS	ISHIDA_E2F1_TARGETS	RAMASWAMY_METASTASIS_DN
BENPORATH_OCCT4_TARGETS	ISHIDA_TARGETS_OF_SYT_SX_FUSIONS	RAMASWAMY_METASTASIS_UP
BENPORATH_PRC2_TARGETS	ISHIKAWA_STING_SIGNALING	RAMIAUN_APOPTOSIS_BY_TGFB1_VIA_MAPK1_DN
BENPORATH_PROLIFERATION	ISSA_EVA_MLL2_TARGETS	RAMIAUN_APOPTOSIS_BY_TGFB1_VIA_MAPK1_UP
BENPORATH_SOX2_TARGETS	ITO_PTTG1_TARGETS_DN	RAMIAUN_APOPTOSIS_BY_TGFB1_VIA_SMAD4_DN
BENPORATH_SUZ12_TARGETS	ITO_PTTG1_TARGETS_UP	RAMIAUN_APOPTOSIS_BY_TGFB1_VIA_SMAD4_UP
BERENJENO_ROCK_SIGNALING_NOT_VIA_RHQA_DN	IVANOVA_MUTATED_IN_COLON_CANCER	RAMPON_ENRICHED_LEARNING_ENVIRONMENT_EARLY_DN
BERENJENO_ROCK_SIGNALING_NOT_VIA_RHQA_UP	IVANOVA_HEMATOPOIESIS_EARLY_PROGENITOR	RAMPON_ENRICHED_LEARNING_ENVIRONMENT_EARLY_UP
BERENJENO_TRANSFORMED_BY_RHOA_DN	IVANOVA_HEMATOPOIESIS_INTERMEDIATE_PROGENITOR	RAMPON_ENRICHED_LEARNING_ENVIRONMENT_LATE_UP
BERENJENO_TRANSFORMED_BY_RHOA_FOREVER_DN	IVANOVA_HEMATOPOIESIS_LATE_PROGENITOR	RANKIN_ANGIOGENIC_TARGETS_OF_VHL_HIF2A_DN
BERENJENO_TRANSFORMED_BY_RHOA_FOREVER_UP	IVANOVA_HEMATOPOIESIS_MATURE_CELL	RANKIN_ANGIOGENIC_TARGETS_OF_VHL_HIF2A_UP
BERENJENO_TRANSFORMED_BY_RHOA_REVERSIBLY_DN	IVANOVA_HEMATOPOIESIS_STEM_CELL	RAO_BOUND_BY_SALL4
BERENJENO_TRANSFORMED_BY_RHOA_REVERSIBLY_UP	IVANOVA_HEMATOPOIESIS_STEM_CELL_AND_PROGENITOR	RAO_BOUND_BY_SALL4_ISOFORM_A
BERENJENO_TRANSFORMED_BY_RHOA_UP	IVANOVA_HEMATOPOIESIS_STEM_CELL_LONG_TERM	RAO_BOUND_BY_SALL4_ISOFORM_B
BERGER_MB02_TARGETS	IVANOVA_HEMATOPOIESIS_STEM_CELL_SHORT_TERM	
BERNARD_PPAPDC1B_TARGETS_DN	IVANOVSKA_MIR106B_TARGETS	
	IVANAGA_CARCINOGENESIS_BY_KRAS_DN	

Journals have dedicated issues to databases



Volume 53, Issue D1

6 January 2025

[Cover image](#)

EISSN 1362-4962

EDITORIAL

MAJOR MULTI-DATABASE
RESOURCES

NUCLEIC ACID SEQUENCE,
STRUCTURE AND REGULATION

PROTEIN SEQUENCE AND
STRUCTURE, MOTIFS AND
DOMAINS

METABOLIC AND SIGNALLING
PATHWAYS, ENZYMES

VIRUSES, BACTERIA, PROTOZOA
AND FUNGI

HUMAN GENOME, MODEL
ORGANISMS, COMPARATIVE
GENOMICS

GENOMIC VARIATION, DISEASES
AND DRUGS

PLANTS

OTHER DATABASES

CORRECTIONS

JOURNAL ARTICLE

CancerSCEM 2.0: an updated data resource of single-cell expression map across various human cancers

Jingyao Zeng , Zhi Nie , Yunfei Shang , Jialin Mai , Yadong Zhang , Yuntian Yang ,
Chenle Xu , Jing Zhao , Zhuojing Fan , Jingfa Xiao ✉ [Author Notes](#)

Nucleic Acids Research, Volume 53, Issue D1, 6 January 2025, Pages D1278–D1286,
<https://doi.org/10.1093/nar/gkae954>

Published: 26 October 2024 **Article history** ▼

JOURNAL ARTICLE

canSAR 2024—an update to the public drug discovery knowledgebase

Phillip W Gingrich , Rezvan Chitsazi , Ansuman Biswas , Chunjie Jiang , Li Zhao ,
Joseph E Tym , Kevin M Brammer , Jun Li , Zhigang Shu , David S Maxwell , Jeffrey A Tacy ,
Ioan L Mica , Michael Darkoh , Patrizio di Micco , Kaitlyn P Russell , Paul Workman ,
Bissan Al-Lazikani ✉

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JOURNAL ARTICLE

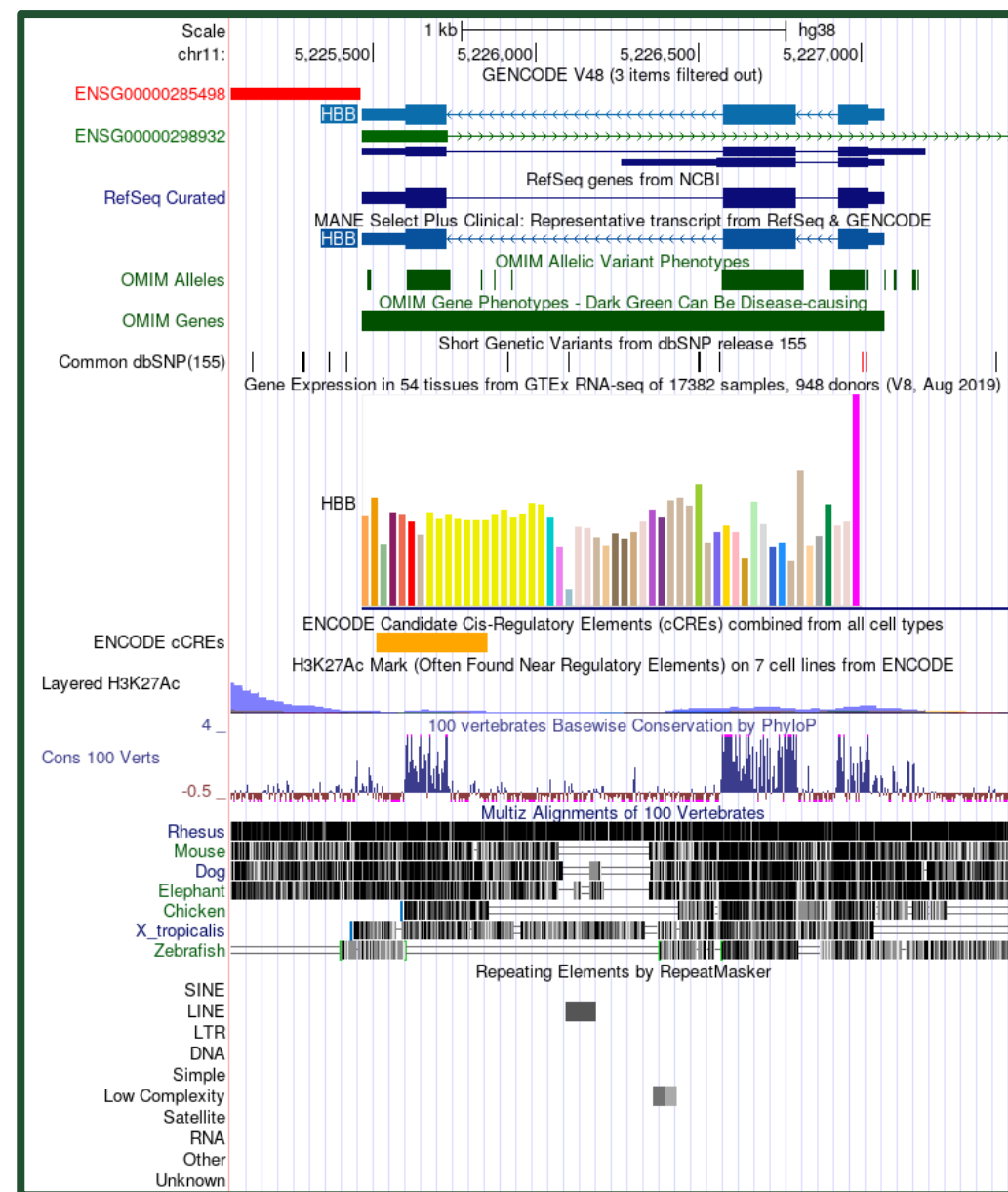
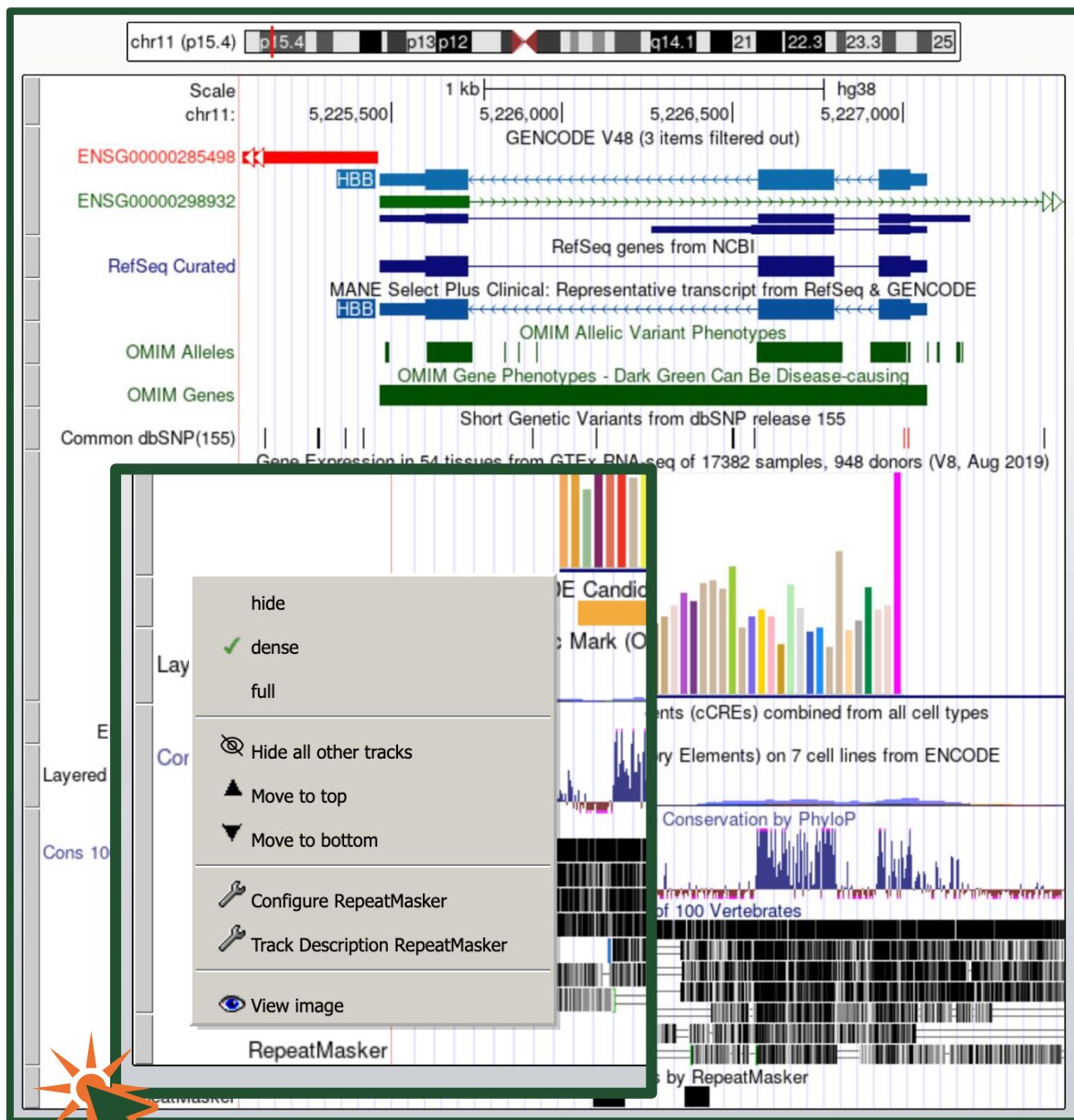
VISTA Enhancer browser: an updated database of tissue-specific developmental enhancers

Michael Kosicki , Fotis A Baltoumas , Guy Kelman , Joshua Boverhof , Yeongshnn Ong ,
Laura E Cook , Diane E Dickel , Georgios A Pavlopoulos , Len A Pennacchio ✉ ,
Axel Visel ✉

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The UCSC Genome Browser contains a rich collection of data tracks



The UCSC Genome Browser contains a rich collection of data tracks

The image shows a screenshot of the UCSC Genome Browser interface. The top navigation bar includes links for Genomes, Genome Browser, Tools, Mirrors, Downloads, My Data, Projects, Help, and About. The main content area is divided into two panels: "COSMIC Track Settings" and "Description".

COSMIC Track Settings

Catalogue of Somatic Mutations in Cancer V101 ([All Phenotypes, Variants, and Literature tracks](#))

Display mode: [Duplicate track](#)

Display data as a density graph: ☐

[Data schema/format description and download](#)

Source data version: COSMIC v101
Assembly: Human Dec. 2013 (GRCh38/hg38)
Data last updated at UCSC: 2025-01-23 23:45:21

Description

[COSMIC](#), the "Catalogue Of Somatic Mutations In Cancer," is an online database of somatic mutations found in human cancer. Focused exclusively on non-inherited acquired mutations, COSMIC combines information from a range of sources, curating the described relationships between cancer phenotypes and gene (and genomic) mutations. These data are then made available in a number of ways including here in the UCSC genome browser, on the COSMIC website with custom analytical tools, or via the [COSMIC sftp server](#). Publications using COSMIC as a data source may cite our reference below.

Genes and Gene Predictions

CRISPR Targets GENCODE Archive GENCODE Versions HGNC Updated IKMC Genes Mapped Non-coding RNA Other RefSeq Pfam in GENCODE Prediction Archive Pseudogenes UniProt

Phenotypes, Variants, and Literature

CADD 1.7 Cancer Gene Expr CIVIC New ClinGen ClinGen CNVs COSMIC Updated COSMIC Regions COVID Data DECIPHER Development Delay GeneReviews GWAS Catalog HGMD public LOVD Variants MITOMAP New [No data-chr11]

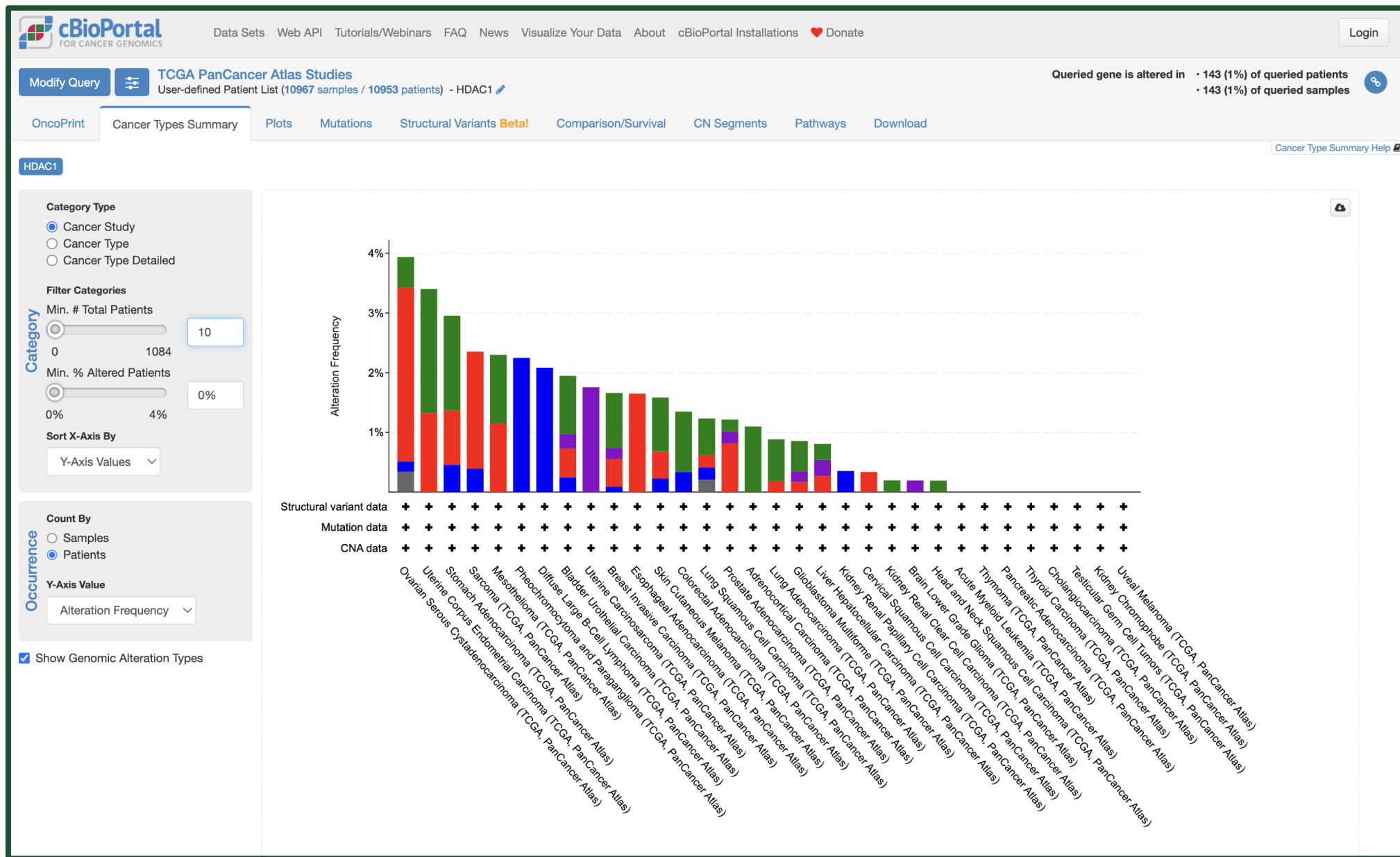
Variation

dbSNP Archive dbVar Common Struct Var DGV Struct Var Genome In a Bottle gnomAD Updated

Human Pangenome - HPRC

Multiple Alignment Pairwise Alignments Rearrangement Short Variants

cBioPortal includes cancer-relevant genomic information



Mutation

Amplification

Deep Deletion

Multiple Alterations

Structural Variant

https://www.cbioportal.org/results/cancerTypesSummary?case_set_id=all&gene_list=HDAC1&cancer_study_list=5c8a7d55e4b046111fee2296

cBioPortal includes cancer-relevant genomic information



[Data Sets](#)
[Web API](#)
[Tutorials/Webinars](#)
[FAQ](#)
[News](#)
[Visualize Your Data](#)
[About](#)
[cBioPortal Installations](#)
[♥ Donate](#)

Login

Modify Query



TCGA PanCancer Atlas Studies
 User-defined Patient List (10967 samples / 10953 patients) - TP53

Queried gene is altered in

• 3939 (36%) of queried patients
 • 3943 (36%) of queried samples

8

OncoPrint

Cancer Types Summary

Plots

Mutations

Structural Variants **Beta!**

Comparison/Survival

CN Segments

Pathways

Download

TP53

Cancer Type Summary Help

Category Type

☒ Cancer Study
☐ Cancer Type
☐ Cancer Type Detailed

Filter Categories

Min. # Total Patients
 10
 0 1084

Min. % Altered Patients
 0%
 0% 92%

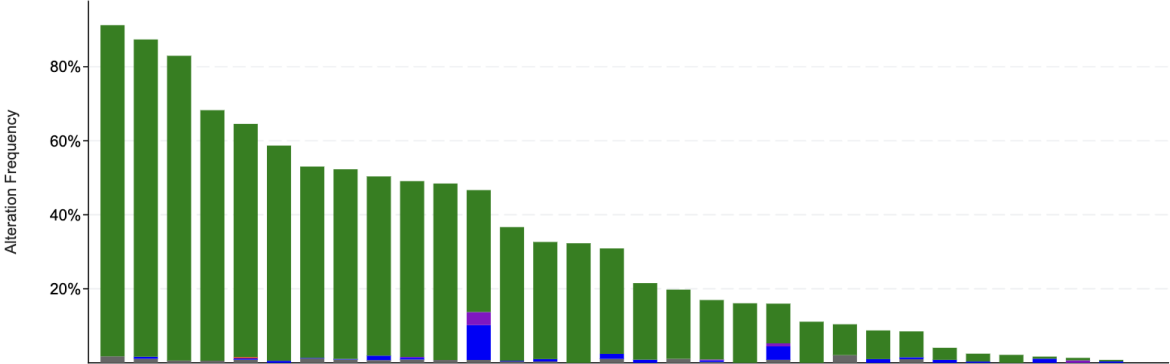
Sort X-Axis By
 Y-Axis Values

Count By

☐ Samples
☒ Patients

Y-Axis Value
 Alteration Frequency

☒ Show Genomic Alteration Types



Cancer Type	Structural variant data	Mutation data	CNA data
Uterine Carcinosarcoma (TCGA, PanCancer Atlas)	+	+	+
Esophageal Adenocarcinoma (TCGA, PanCancer Atlas)	+	+	+
Lung Squamous Cell Carcinoma (TCGA, PanCancer Atlas)	+	+	+
Head and Neck Squamous Cell Carcinoma (TCGA, PanCancer Atlas)	+	+	+
Ovarian Serous Cystadenocarcinoma (TCGA, PanCancer Atlas)	+	+	+
Pancreatic Adenocarcinoma (TCGA, PanCancer Atlas)	+	+	+
Colorectal Adenocarcinoma (TCGA, PanCancer Atlas)	+	+	+
Lung Adenocarcinoma (TCGA, PanCancer Atlas)	+	+	+
Bladder Urothelial Carcinoma (TCGA, PanCancer Atlas)	+	+	+
Stomach Adenocarcinoma (TCGA, PanCancer Atlas)	+	+	+
Brain Lower Grade Glioma (TCGA, PanCancer Atlas)	+	+	+
Sarcoma (TCGA, PanCancer Atlas)	+	+	+
Uterine Endometrial Carcinoma (TCGA, PanCancer Atlas)	+	+	+
Breast Invasive Carcinoma (TCGA, PanCancer Atlas)	+	+	+
Kidney Chromophobe (TCGA, PanCancer Atlas)	+	+	+
Liver Hepatocellular Carcinoma (TCGA, PanCancer Atlas)	+	+	+
Glioblastoma (TCGA, PanCancer Atlas)	+	+	+
Adrenocortical Carcinoma (TCGA, PanCancer Atlas)	+	+	+
Skin Cutaneous Melanoma (TCGA, PanCancer Atlas)	+	+	+
Mesothelioma (TCGA, PanCancer Atlas)	+	+	+
Prostate Adenocarcinoma (TCGA, PanCancer Atlas)	+	+	+
Cholangiocarcinoma (TCGA, PanCancer Atlas)	+	+	+
Diffuse Large B-Cell Lymphoma (TCGA, PanCancer Atlas)	+	+	+
Cervical Squamous Cell Carcinoma (TCGA, PanCancer Atlas)	+	+	+
Acute Myeloid Leukemia (TCGA, PanCancer Atlas)	+	+	+
Thymoma (TCGA, PanCancer Atlas)	+	+	+
Kidney Renal Papillary Cell Carcinoma (TCGA, PanCancer Atlas)	+	+	+
Kidney Renal Clear Cell Carcinoma (TCGA, PanCancer Atlas)	+	+	+
Pheochromocytoma and Paraganglioma (TCGA, PanCancer Atlas)	+	+	+
Testicular Germ Cell Tumors (TCGA, PanCancer Atlas)	+	+	+
Thyroid Carcinoma (TCGA, PanCancer Atlas)	+	+	+
Uveal Melanoma (TCGA, PanCancer Atlas)	+	+	+

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Cancer Types Summary
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Groups: (drag to reorder)
Altered group (3943/3939)
Unaltered group (7017/7011)
Unprofiled group (7)
TP53 (3943/3939)
Select all | Deselect all

Overlap
Survival
Clinical
Genomic Alterations
mRNA
Protein
Arm-level CNA
Genetic Ancestry
Methylation
Phosphosite Quantification

Interpret all results with caution, as they can be confounded by many different variables that are not controlled for in these analyses. Consider consulting a statistician.

The log-rank test is used to test the null hypothesis that there is no difference between the groups in the probability of an event at any time point. Hazard ratios are derived from the log-rank test.

The survival data on patients from different cohorts may have been defined by [different criteria](#).

Min. # Patients per Group:
Columns:

Survival Type	Number of Patients	p-Value	q-Value
Overall	10803	< 10 ⁻¹⁰	< 10 ⁻¹⁰
Disease Free	5383	< 10 ⁻¹⁰	< 10 ⁻¹⁰
Disease-specific	10258	< 10 ⁻¹⁰	< 10 ⁻¹⁰
Progression Free	10613	< 10 ⁻¹⁰	< 10 ⁻¹⁰

Showing 1-4 of 4

Overall

☐ Calculate hazard ratios

Altered group

☐ Add landmarks

Add landmark values

X-Axis Max:
 Months Survival

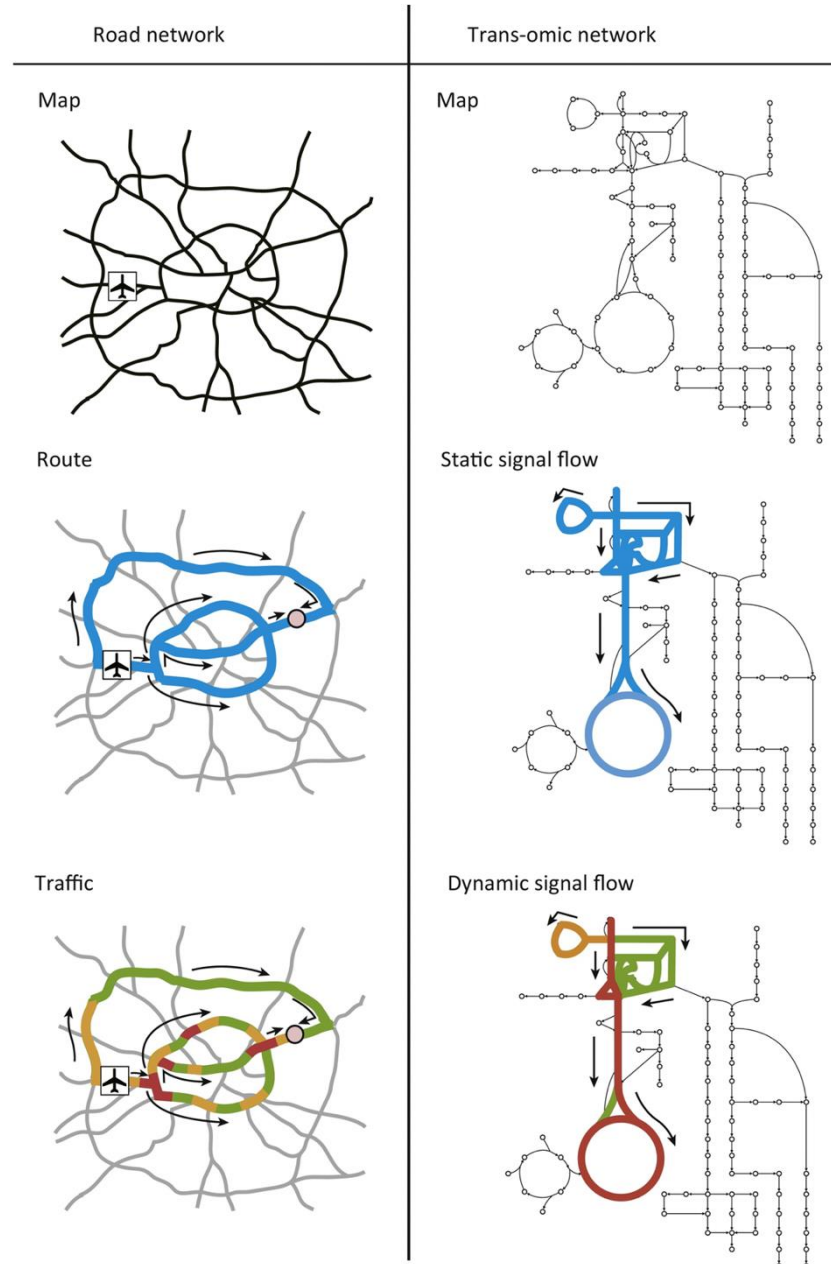
Logrank Test P-Value: 0.00

- Altered group
- Unaltered group

	0	20	40	60	80	100	120	140	160	180	200	220	240	260	280	300	320	340	360
Altered group	3890	2069	980	539	287	164	84	45	23	14	12	9	5	3	3	1	0	0	0
Unaltered group	6913	4034	2127	1219	710	393	229	130	86	53	41	25	13	9	6	5	4	4	2

Survival plot summary

	Number of Cases, Total	Number of Events	Median Months Overall (95% CI)
Altered group	3890	1480	55.20 (51.16 - 60.66)
Unaltered group	6913	2033	95.34 (89.16 - 105.04)



Genome

What reactions are able to happen under any physiological conditions

Epigenome, Transcriptome

What reactions seem to be active under specific physiological conditions

Proteome, Metabolome

What reactions are active under specific physiological conditions

Genome organization: from biology to computer bytes



Gene = exon(s) + intron(s)

22 autosomal chromosomes: 1, 2,.. 22
1 pair of sex chromosomes: X, Y
1 mitochondrial chromosome: M or MT

Example: “gene001” starts at position 10 and ends at position 58.

It has 3 exons, at positions: 10-20, 34-48, 55-58

It has 2 introns, at positions: 21-33, 49-54

Let’s assume, this gene has three transcript variants:

“transcript001”: Exon1 + Exon2 + Exon3

“transcript002”: Exon2 + Exon3 (different transcription start site)

“transcript003”: Exon1 + Exon3 (alternative splicing)

Genomic coordinates:

Gene001: 10-58

Transcript001: 10-58

Transcript002: 34-58

Transcript003: 10-58

In BED format (chromosome, start, end, annotation):

chr1 10 58 gene001

chr1 10 58 transcript001|gene001

chr1 34 58 transcript002|gene001

chr1 10 58 transcript003|gene001

<https://genome.ucsc.edu/FAQ/FAQformat.html#format1>

- Go to www.ensembl.org
- Click on “BioMart” at the top of the page
- Choose “Ensembl genes 114” as the database
- Choose “Human genes (GRCh38.p14)” as the dataset.
 - From this page you can access all human genes.
 - Clicking on “Attributes” you can choose what information you want to see.
 - Clicking on “Filtering” you can filter the information.

We will use BioMart, a tool that allows easy access of genomic data.
More info: <http://useast.ensembl.org/info/data/biomart/index.html>

-- Our goal --

Get a file with the following information (attributes):

- genomic location of the transcript
- transcript ID and type
- Gene ID, name and type

Show only the genes (filtering):

- on primary chromosomes

Working with BioMart

The screenshot shows the Ensembl BioMart interface. Callout 1 points to the 'Filters' section in the left sidebar. Callout 2 points to the 'Attributes' section in the left sidebar. Callout 3 points to the 'Results' button in the top navigation bar. Callout 4 points to the 'Export' dropdown menu. Callout 5 points to the 'Go' button next to the export options.

Dataset
Human genes (GRCh38.p14)

Filters
Chromosome/scaffold: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, MT, X, Y

Attributes
Chromosome/scaffold name
Transcript start (bp)
Transcript end (bp)
Transcript stable ID
Transcript type
Gene stable ID
Gene name
Gene type

Dataset
[None Selected]

Export all results to: Compressed file (.gz) TSV ☐ Unique results only **Go**

Email notification to:

View: 10 rows as HTML ☐ Unique results only

Chromosome/scaffold name	Transcript start (bp)	Transcript end (bp)	Transcript stable ID	Transcript type	Gene stable ID	Gene name	Gene type
1	3069168	3434342	ENST00000511072	protein_coding	ENSG00000142611	PRDM16	protein_coding
1	3069183	3186591	ENST00000607632	retained_intron	ENSG00000142611	PRDM16	protein_coding
1	3069197	3435421	ENST00000378391	protein_coding	ENSG00000142611	PRDM16	protein_coding
1	3069211	3434342	ENST00000514189	protein_coding	ENSG00000142611	PRDM16	protein_coding
1	3069203	3438621	ENST00000270722	protein_coding	ENSG00000142611	PRDM16	protein_coding
1	3237931	3433925	ENST00000512462	protein_coding_CDS_not_defined	ENSG00000142611	PRDM16	protein_coding
1	3244132	3386918	ENST00000463591	protein_coding	ENSG00000142611	PRDM16	protein_coding
1	3396491	3434293	ENST00000509860	protein_coding	ENSG00000142611	PRDM16	protein_coding
1	3424919	3434095	ENST00000378389	protein_coding_CDS_not_defined	ENSG00000142611	PRDM16	protein_coding
1	3425216	3426072	ENST00000606170	retained_intron	ENSG00000142611	PRDM16	protein_coding

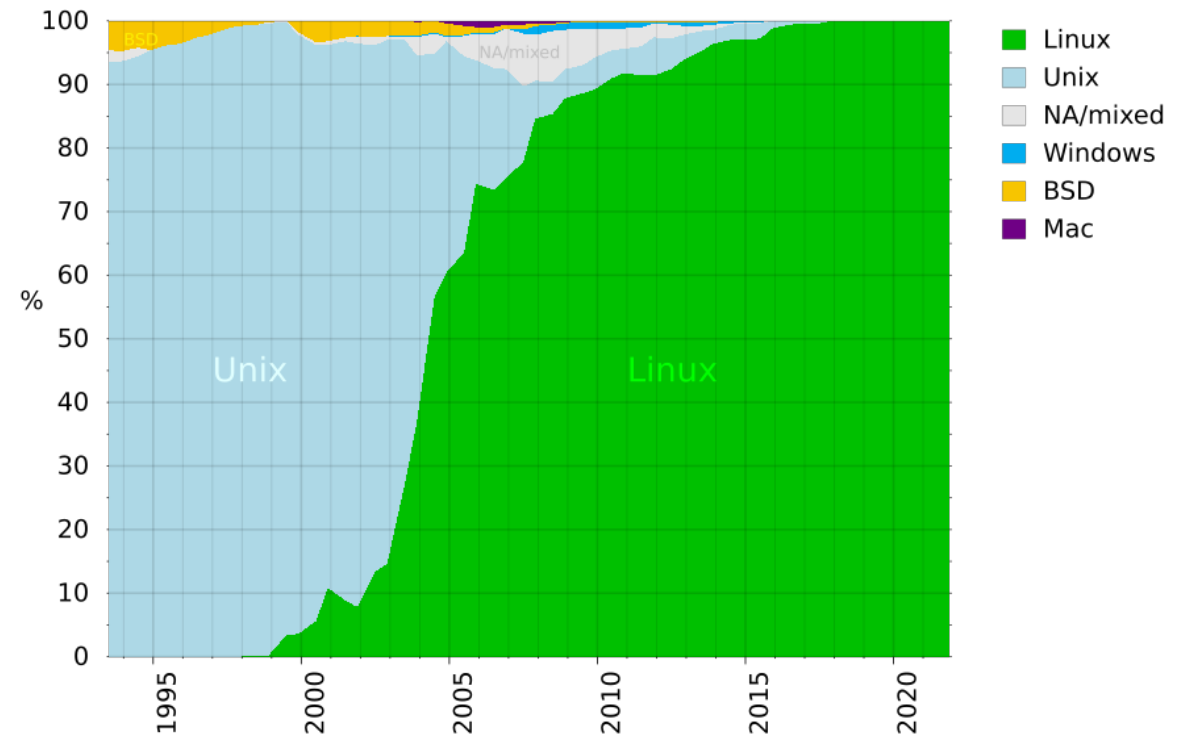
1. Click “Filters” and select these chromosomes (you will find them under “REGION”)
2. Click “Attributes” and select these ones in **this specific order** (you will find them under “GENE”)
3. Click “Results”
4. Select “Compressed file (.gz)”
5. Click “Go” and save the file in the directory you will be working (e.g. make a folder “workshop” on your desktop)

Questions we can answer using the file from biomart:

- Which chromosome has the highest number of protein coding genes?
- Which protein-coding gene has the most coding splice variants?
- Which protein-coding gene has the most non-coding splice variants?

Which tool can we use?

Linux / bash



Operating systems used on top 500 supercomputers

By Benedikt.Seidl - Eigenes Werk (own work) / source top500.org, Public Domain

https://en.m.wikipedia.org/wiki/File:Operating_systems_used_on_top_500_supercomputers.svg

Working with linux / bash

Unix is a command-line programming language.

Instead of clicking, you type commands. For example, when you want to make a font bold, then you click on the “B” button. This click is translated to the command “make the selected text bold”. In unix you would need to type this.

Open the terminal, type “bash” and hit “return” (or “enter”). This will get you into the bash environment (the default may be zsh, which uses slightly different commands and syntax).

Type “pwd” and hit return. This will tell you in which directory you currently are.

If you made the folder “workshop” on your desktop, you can move to it with the “cd” command.

This is what I got when running the above (the orange ones are what I typed):

```
Last login: Sun Jun 01 14:30:17 on ttys001

The default interactive shell is now zsh.
To update your account to use zsh, please run `chsh -s /bin/zsh`.
For more details, please visit https://support.apple.com/kb/HT208050.
UH-GDG7LR6RT7:~ axt5207$ pwd
/Users/axt5207
UH-GDG7LR6RT7:~ axt5207$ cd Desktop/workshop/
UH-GDG7LR6RT7:workshop axt5207$ pwd
/Users/axt5207/Desktop/workshop
UH-GDG7LR6RT7:workshop axt5207$ ls
mart_export.txt.gz
UH-GDG7LR6RT7:workshop axt5207$
```

To see the manual of a specific command, e.g. of pwd, type “man pwd” hit return. Scroll up/down with arrows, hit “q” to exit the manual page. Read the manual pages of the commands you use!

An online tutorial: <https://linuxcommand.org/>

We should be ready to
start with **exercise 1**

**The most commonly used
unix commands**

cat	mv
cd	paste
colrm	pwd
cut	rm
cp	sed
exit	sort
ftp	scp
grep	ssh
head	sftp
join	tail
less	uniq
ls	xterm
man	wc
more	wget

